

Ritam Koley

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PROFESSIONAL SUMMARY

I am a Computer Science undergraduate at NIT Durgapur with profound expertise in competitive programming, algorithms, and cybersecurity. My core technical interests are focused on advancing in Artificial Intelligence and Machine Learning. Alongside my engineering pursuits, I have a strong creative passion for graphic design and video editing. Outside of academics and coding, my hobbies include playing football, reading books, and hitting the gym, which help me maintain a disciplined, highly analytical, and well-rounded approach to both my personal and professional life.

EDUCATION

National Institute of Technology, Durgapur B.Tech in Computer Science <i>Coursework:</i> Computer Networks, Operating System, Algorithms, Database Management, Object-Oriented Programming, Artificial Intelligence	Aug 2023 – Present CGPA: 9.1/10
DAV Model School, Durgapur Class XII (Higher Secondary Examination) - CBSE Board	2021 – 2023 Percentage: 96.8%
St. Xavier's School, Durgapur Class X (Secondary Examination) - CISCE Board	2009 – 2021 Percentage: 95.6%

PROFESSIONAL EXPERIENCE

Goldman Sachs <i>Engineer Analyst, Investment Banking Department</i> - Engineered automated migration pipelines to seamlessly transfer operational dashboards and system metrics from Datadog to AWS CloudWatch. - Designed the observability infrastructure for the Global Banking and Markets-Private and Global Investment Research Divisions. - Collaborated cross-functionally with the AWS team to execute the automation strategy and ensure high-availability system monitoring.	May 2026 – Jul 2026
IIT Delhi <i>Research Intern</i> - Developed an automated security testing tool in Python to identify malicious VPNs and browser extensions prone to unauthorized data leakage. - Implemented network sniffing and traffic analysis scripts to detect IP leaks, DNS leaks, and unauthorized routing to malicious locations. - Evaluated extensions for systemic vulnerabilities that expose sensitive user search history and browsing data.	Oct 2025 – Jan 2026
IIT Kanpur <i>Research Intern</i> - Researched and implemented Secured Multiparty Computation and Fluid Multiparty Computation. - Created a custom cryptographic library in Rust to execute Secure Multiparty Computation protocols. - Acquired practical knowledge in advanced security concepts including Secret Sharing and Post-Quantum Cryptography.	May 2025 – Jul 2025
Physics Wallah <i>Mentor</i> - Mentored and guided JEE aspirants by providing strategic preparation insights and conceptual clarity. - Conducted regular doubt-solving sessions to resolve complex academic queries and improve student problem-solving speeds.	Aug 2024 – Oct 2024

PROJECTS

FlipGears (Flipkart Grid 6.0 Project) <i>FastAPI, TailwindCSS, ReactJS, JavaScript</i> - Developed a vision-based system leveraging machine learning and image processing to detect the freshness of fruits and vegetables. - Implemented OCR techniques to automatically read printed expiration dates on packaged products for precise expiry detection. - Integrated a deep learning model to accurately identify product brands via packaging and logo recognition. - Built a responsive frontend interface for uploading images, displaying analysis results, and generating bills.	GitHub Link
AQI-Level Predictor <i>Python, Pandas, Matplotlib, Scikit-learn</i> - Designed a predictive model utilizing historical data and trends to forecast Air Quality Index (AQI) levels for the next 50 years. - Applied regression-based analysis on AQI level graphs to accurately estimate the future environmental conditions of a city. - Implemented a proactive warning system that triggers immediate alerts if predicted AQI values exceed hazardous environmental thresholds.	GitHub Link
KeyLogger <i>Python, Pynput</i> - Developed a Python-based key-logger to reliably capture and record all keystrokes typed on the keyboard in real time. - Enabled the secure logging of sensitive inputs such as usernames and passwords, explicitly tailored for ethical hacking and penetration testing use cases. - Demonstrated a deep understanding of low-level input monitoring and event handling mechanics within operating systems.	GitHub Link

ACHIEVEMENTS

- Solved 1000+ Data Structures and Algorithms problems combined across platforms including LeetCode, CodeChef, and GeeksforGeeks.
- Achieved LeetCode Guardian status with peak rating of 2141 and secured 208th global rank in LeetCode Biweekly Contest 170 with all-kill tag.
- Secured 23rd global rank in CodeChef Starters 235 and ranked as a 4-star coder on CodeChef.
- Secured Pre-finalist standing (Top 80 nationwide) in the Flipkart Grid 6.0 Software Development Track.
- Ranked in the top 1% in the JEE Main and JEE Advanced examinations (2023 and 2024).

TECHNICAL SKILLS

- Languages:** C, C++, Python, Rust, HTML, CSS, JavaScript, TypeScript, ShellScript.
- Technologies:** ReactJS, NodeJS, ExpressJS, Redis, MongoDB, PostgreSQL, MySQL, TailwindCSS, ThreeJS, Figma, Spline.
- Cloud Technologies:** AWS CloudWatch, AWS Lambda, AWS EC2, GCP, Datadog.
- DevOps Tools:** Jenkins, Docker, Kubernetes, Terraform.
- Cybersecurity Tools:** BurpSuite, Wireshark, Metasploit, NMAP, John the Ripper.
- Soft Skills:** Leadership, Time Management, Communication.